



Reflexive AI

REFLEXIVE AI RESEARCH INITIATIVE

Human-Centered AI

Overview of relevant insights from scientific research

10 minutes

details & sources @ <https://github.com/LucidInstitute/Reflexive-AI-Initiative/tree/main/Literature>

DEFINING AI

AI can take over human thinking, and technology is never neutral.

// KEY FINDINGS

- 1 OECD definition.** A machine-based system that infers from input how to generate predictions, content, recommendations or decisions (OECD, 2024). More simply: “Computers that do the sorts of things minds can do” (Boden, 2016).
- 2 Minds, delegated.** Unlike earlier revolutions, thinking and creativity can now be delegated, not only physical action. (Boden, 2016; Hilbert, 2020). Robotics will also replace more physical action.
- 3 Technology embodies power.** It shapes how we see the world and imposes behaviors on people. (Winner, 1985; Latour, 1992)
- 4 Bias carries over.** AI can perpetuate biases already present in its training data. (Barocas & Selbst, 2016; Noble, 2018)

Individual gains are real, but they do not automatically become team gains.

// KEY FINDINGS

1

Firm level. Positive productivity effects across sectors, stronger when AI is integrated into work activities. (Necula et al., 2024; Czarnitzki et al., 2023)

2

Employee level. In a survey of 359 bank employees, AI use raised productivity and engagement. (Gusti et al., 2024)

3

Perception gap. Developers felt more productive, yet commit activity did not change and experienced developers were not faster. (Stray et al., 2026; Becker et al., 2025)

4

Lost downstream. Local gains are absorbed by workflow and organizational constraints. (DORA, 2025; Russo, 2026)

Whether AI helps or harms depends on trust, fit and culture.

// KEY FINDINGS

1

Engagement. AI raised engagement in two manufacturers, unless self-efficacy is low or jobs were cut. (Tortorella et al., 2025)

2

Trust. Low employer trust sharply reduces engagement; AI may bring an “alienational” contract. (Braganza et al., 2021)

3

Preconditions. Task fit, transparency and expertise. Black boxes breed aversion; bureaucracy builds “algorithmic cages”. (Bankins et al., 2023; Meijer et al., 2021)

4

Six displacements. Software engineers report epistemic, identity, normative, phenomenological, social and meaning shifts. (Tavantzis et al., in press)

AI can amplify existing inequality unless we design purposefully against it.

// KEY FINDINGS

- 1 Bias in practice.** A widely used health system under-recommended treatment for black relative to white patients. (Obermeyer et al., 2019)
- 2 Hard to remove.** Mitigation strategies exist, but implicit biases are difficult to eradicate. (Nazer et al., 2023; Howard & Borenstein, 2017)
- 3 Digital divide.** Inequality can grow between households that can use AI and those that cannot. (Cornelli et al., 2023)
- 4 Labor market.** Automation can replace low-skilled tasks, leaving some workers worse off despite seeming economic growth. (Agrawal et al., 2019; Li & Nichols, 2026)

Human-centered AI balances human control and automation, guided by shared values.

// KEY FINDINGS

- 1 One scale is not enough.** Early models run from no automation to full autonomy, as in self-driving cars, and force a choice. Too much control or too much automation can both add risk. (Sheridan & Verplank, 1978; SAE, 2014)
- 2 Two dimensions.** Human control and computer automation are separate axes. Life-critical systems such as pacemakers and self-driving cars need both high to be reliable, safe and trustworthy. (Shneiderman, 2020)
- 3 Copenhagen Manifesto.** 35 scholars set five values and twelve principles on human dignity, autonomy and collective wellbeing, including human sovereignty and responsible management. (Russo et al., 2024)
- 4 Working definition.** No agreed definition yet. Our draft: efficient work while people stay in control, motivated, connected, and free to question, change or reject AI decisions. (Initiative draft)

Governance must turn principles into structures, processes and enforcement.

// WHERE TO FOCUS

- 1 Answer how, not just what.** Ethical guidelines say what, not how, and can lose to commercial interests. Define structures, processes and relational mechanisms, with enforcement. (Mittelstadt, 2019; Mäntymäki et al., 2022)
- 2 Cover who, what, when and how.** Define who governs, what is governed, when across the lifecycle, and how through policies, tools and models. Think system-level, not just algorithm ethics. (Batool et al., 2025; Lu et al., 2022)
- 3 Go beyond fairness and privacy.** Most solutions address only one or two principles. Build in the human-centered ones: control, autonomy, wellbeing and creativity. (Batool et al., 2025; Russo et al., 2024)
- 4 Make it actionable.** Use concrete patterns, fit them to culture, leadership and up-skilling, and test with Action Research in real organizations. (Lu et al., 2022; Tavantzis et al., 2027)

Sixteen starting themes for human-centered AI

DIRECTION & CONTROL

Purpose & Strategy ^{T·K}

What does human-centered mean here, and which real need does each AI use serve?

Human Agency ^{T·K}

Who decides, how far may AI agents act alone, and who answers for the result?

Governance & Ethics ^{T·K}

Which rules, roles and enforcement turn our principles into daily practice?

Review & Adaptation ^{T·K}

How do we track outcomes, harms and wellbeing, and adapt at a human pace?

PEOPLE & WORK

Meaningful Work ^K

Which tasks must stay human to keep work competent and meaningful?

Work Design ^{T·K}

How do we split tasks between people and AI and redesign workflows?

Workforce & Learning ^{T·K}

Which skills must people build or keep to use, verify and question AI?

Quality & Oversight ^{T·K}

How do we verify AI output, surface uncertainty and prevent over-reliance?

CULTURE & VOICE

Communication ^{T·K}

Do people know where, how and why AI is used, and how their data is handled?

Participation ^{T·K}

How do employees and their representatives co-design and contest AI use?

Questioning Culture ^{T·K}

Can people experiment, make mistakes and reject AI output without pressure?

Power & Dynamics ^T

Whose interests, roles and timing shape AI, and where does resistance come from?

BOUNDARIES & DUTIES

Risk & Boundaries ^{T·K}

Where is AI restricted or off-limits, e.g. HR, health, monitoring of staff?

Legal & Compliance ^{T·K}

Which laws and regulations, like the EU AI Act, apply to each AI use?

IP & Data Ownership ^T

Who owns AI inputs, outputs and data, and what may leave the organization?

Sustainability ^K

What is the environmental and long-term societal cost of our AI use?

Twelve starting themes for AI governance.

DIRECTION & ACCOUNTABILITY

AI Strategy Design

Anchor AI goals in purpose and human-centered aims

AI Strategy Evaluation

Review outcomes and harms, not only productivity

Leadership

Set ownership and accountability for AI decisions

Governance & Ethics

Turn principles into rules, roles and enforcement

PEOPLE & ORGANIZATION

Collaboration

Involve employees, IT, HR, legal and risk in decisions

Communication

Keep AI use and decision logic transparent

Organizational Culture

Make it safe to question or reject AI output

Organizational Dynamics

Account for power, roles and workflow effects

CAPABILITY & OBLIGATIONS

Workforce Development / HRM

Up-skill people; guard fairness when HR uses AI

Sustainability

Weigh environmental and long-term societal impact

Legal

Map regulatory and compliance duties

Intellectual property & ownership

Clarify who owns AI inputs, outputs and data

THANK YOU

**Now its up to us to create concrete
help for organizations**